Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

Hao-Chun Hsieh **Pao-Hua Hsu** **Nien-Tse Wu**
Institute of Artificial Intelligence Innovation
National Yang Ming Chiao Tung University
{handsome.ii14, paohuah.ii14, mattwu.ii14}@nycu.edu.tw

1 Project Overview

1.1 Profile

Track: 3. Application

Abstract and project goal: We address the problem of emotional capability loss when an expressive TTS model trained on a high-resource language is adapted to a low-resource language. Although the original model can generate emotionally expressive speech, its emotional controllability and prosodic richness often degrade after fine-tuning on limited target-language data. To tackle this issue, we propose an RL-enhanced adaptation framework built upon IndexTTS2 to preserve and realign emotional expressiveness during low-resource language transfer [Zhou et al., 2025]. Our method aims not to learn emotion from scratch, but to retain the emotional prior of the pretrained expressive model while improving emotion intensity, emphasis control, and speech naturalness in the low-resource language. In our study, we instantiate this setting using Chinese as the high-resource source language and Taiwanese as the low-resource target language.

TL;DR (“Too Long; Didn’t Read"): This project studies how to preserve and recover the emotional expressiveness of an expressive TTS model when adapting it from a high-resource language to a low-resource language, using reinforcement learning to improve emotion retention and prosodic control.

1.2 Motivation

Expressive speech synthesis is increasingly important for applications such as dubbing, broadcasting, and interactive media. However, when a strong expressive TTS model trained on a high-resource language is adapted to a low-resource language, the model often retains intelligibility but loses emotional expressiveness and fine-grained prosodic control. This makes low-resource adaptation not only a language transfer problem, but also an emotion preservation and alignment problem.

- **Why is the problem interesting?** This problem is interesting because it lies at the intersection of low-resource language adaptation, cross-lingual transfer, and controllable expressive speech synthesis. Rather than learning emotional speech generation from scratch, we focus on preserving and realigning the expressive capability that already exists in the pretrained TTS model when transferring it to a low-resource language.
- **Challenge 1: Emotional Forgetting During Low-Resource Adaptation.** When an emotionally expressive TTS model trained on a high-resource language is fine-tuned on limited low-resource language speech data, it often suffers from catastrophic forgetting, causing the synthesized speech to become flatter and less expressive.

- **Challenge 2: Loss of Fine-Grained Prosodic Control.** After low-resource adaptation, the model may still produce intelligible speech, but fail to preserve natural rhythm, pause structure, emphasis, and emotional intensity that are essential for expressive speech generation.
- **Challenge 3: Difficulty of Cross-Lingual Emotion Alignment.** Emotional cues and prosodic patterns do not transfer perfectly across languages. It is therefore difficult to preserve the expressive prior learned from a high-resource language and align it with the pronunciation, prosody, and linguistic structure of a low-resource language using only limited target-language data.
- **Why the problem remains open or unsolved?** This problem remains open because low-resource languages lack sufficient emotionally annotated speech data for conventional supervised training. As a result, naive fine-tuning often prioritizes pronunciation adaptation over expressive preservation. Existing TTS systems can achieve strong intelligibility in cross-lingual or low-resource settings, but maintaining emotional controllability and prosodic richness during adaptation is still challenging. Relevant prior works attempting to address speech emotion and expressiveness include IndexTTS2 [Zhou et al., 2025], emotion2vec [Ma et al., 2023], and other expressive or controllable speech synthesis systems.
- **State-of-the-Art (SOTA) Method:** Currently, IndexTTS2 provides a strong foundation for zero-shot and controllable speech synthesis [Zhou et al., 2025]. In this project, we use it as the expressive base model and investigate how reinforcement learning can help preserve and realign its emotional capabilities during adaptation from a high-resource language to a low-resource language.

2 Problem Formulation

We formulate emotional preservation and prosodic realignment during low-resource language adaptation as a fully observable Markov Decision Process (MDP). Our setting starts from an expressive TTS model pretrained on a high-resource language, and considers the adaptation process to a low-resource target language. Within the IndexTTS2 framework, the environment is defined by the autoregressive Text-to-Semantic (T2S) generation process, where the agent (the language model policy π_θ) generates a sequence of discrete semantic speech tokens token-by-token. At any step, the state s encompasses the available conditioning information: the input text sequence, a speaker timbre embedding c, a duration constraint embedding p, and a target global emotion embedding e. The action a represents the generated semantic token sequence, which is subsequently converted into a mel-spectrogram by the Semantic-to-Mel (S2M) flow-matching module. Although the policy operates on semantic tokens, the reward is computed from the decoded speech waveform.

Our objective is not to learn emotional speech generation from scratch, but to preserve and recover the emotional prior of the pretrained expressive model after low-resource adaptation. To reduce severe forgetting during transfer, we first assume a parameter-efficient adaptation stage that aligns the model to the target language while keeping most expressive knowledge intact. On top of this adapted model, the RL learner receives a multi-objective scalar reward signal $R(s, a)$ upon generating a complete sequence. This feedback is composed of four task-specific components: an Emotion Classification Reward to ensure categorical accuracy, a Global Emotion Intensity Reward measured by the distance to a neutral centroid in the Valence-Arousal-Dominance (VAD) space, a Local Emphasis Control Reward calculated from the relative pitch (f_{pitch}) and energy (f_{energy}) patterns of target emphasized words, and an ASR-Based Content Reward to preserve intelligibility and semantic correctness in the target language. The sequence-level reward can therefore be written as

$$R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}},$$

where the coefficients λ_i balance emotional accuracy, global intensity, local prosody, and content preservation.

The optimization problem seeks to maximize the expected reward using Group Relative Policy Optimization (GRPO). Specifically, the objective function aims to optimize the policy parameters θ by maximizing $\mathbb{E}_{s\sim\mathcal{D},a\sim\pi_\theta}[R(s, a)\nabla_\theta \log \pi_\theta(a|s)]$, subject to a Kullback-Leibler (KL) divergence constraint $\beta\text{KL}(\pi_\theta(\cdot|s)\|p_{\text{adapt}}(\cdot|s))$ to prevent catastrophic deviation from the adapted base model and to preserve its pretrained expressive prior. We operate under the assumption that the discrete semantic token space produced by the T2S module is sufficiently expressive to retain and recover subtle emotional intensities and local emphasis variations prior to acoustic reconstruction, even after transfer from a high-resource language to a low-resource language.

3 Empirical Evaluation

To rigorously evaluate the effectiveness of our proposed RL-enhanced IndexTTS2 framework in preserving and recovering emotional expressiveness during transfer from a high-resource language to a low-resource language, we employ a combination of objective and subjective evaluations.

3.1 Performance Metrics

We plan to adopt several key performance metrics to comprehensively assess speech intelligibility, speaker fidelity, emotional expressiveness, and emotion retention after low-resource adaptation.

- **Objective Metrics**:
  - **Word Error Rate (WER)**: Evaluates speech intelligibility and articulation accuracy using ASR models like FunASR or Whisper [Radford et al., 2022].
  - **Speaker Similarity (SS)**: Computed as the cosine similarity between speaker embeddings extracted from a pretrained speaker recognition model to ensure the timbre is maintained after low-resource adaptation and RL optimization.
  - **Emotion Similarity (ES) / Accuracy**: Measured using the open-source emotion2vec model to objectively verify if the generated speech matches the target emotion category.
  - **Emotion Retention Gap (ERG)**: Let M denote an emotion-related metric such as Emotion Similarity or Emotion MOS. We define $\mathrm{ERG} = (M_{\text{base}} - M_{\text{adapt}})/M_{\text{base}}$ to quantify the relative emotional degradation after low-resource adaptation, and additionally report the recovery ratio $(M_{\text{ours}} - M_{\text{adapt}})/(M_{\text{base}} - M_{\text{adapt}})$ to measure how much of the lost emotional capability is restored by our method.
- **Subjective Metrics (MOS Framework)**:
  - **Multidimensional MOS**: We will conduct human evaluations based on a 1-5 scale for Quality MOS (QMOS), Similarity MOS (SMOS), Prosody MOS (PMOS), and Emotion MOS (EMOS).
  - **Emotion Intensity Test (EIT)**: A pairwise comparison task where listeners identify the stronger emotional sample among weak, medium, and strong synthesized speech.
  - **Emphasis Accuracy Test (EAT)**: Assesses the consistency between the model's predicted emphasis positions (local prosody) and those perceived by human listeners.

The combination of EIT and EAT is crucial as our RL method explicitly optimizes for fine-grained intensity and local emphasis, which standard MOS cannot fully capture.

3.2 Baseline Methods

We compare our method against several baselines to explicitly quantify emotional forgetting and emotional recovery during low-resource adaptation.

- **Original High-Resource IndexTTS2**: The expressive source model before low-resource adaptation, used to establish the upper bound of emotional capability prior to transfer.
- **IndexTTS2 + Naive Low-Resource Fine-Tuning**: A baseline that fine-tunes IndexTTS2 on limited target-language data without any explicit preservation mechanism, used to measure the severity of emotional forgetting.
- **IndexTTS2 + Parameter-Efficient Adaptation**: A target-language adaptation baseline using a lightweight adaptation strategy, used to test whether preserving more pretrained parameters helps retain emotional expressiveness.
- **IndexTTS2 + Parameter-Efficient Adaptation + RL (Ours)**: Our full method, which applies RL-based emotional and prosodic alignment after low-resource adaptation.
- **CosyVoice2** [Du et al., 2024]: A strong zero-shot text-to-speech baseline with predefined emotion instructions for cross-model comparison.

3.3 Benchmark Tasks or Datasets

To simulate the low-resource fine-tuning scenario and evaluate catastrophic forgetting of emotions, we utilize the following datasets:

- **High-Resource Pre-training Data**: The expressive base model is first trained on large-scale high-resource speech corpora to establish strong semantic generation and emotional priors.
- **Emotion Supervision Data**: We utilize the Emotional Speech Database (ESD) and the Expresso dataset, which provide annotations for discrete emotions, continuous intensities, and emphasis.
- **Low-Resource Adaptation Task**: We construct a target-language adaptation task by taking a small subset (e.g., 1 to 5 hours) of a low-resource language corpus. In our study, this setting is instantiated with Chinese as the high-resource source language and Taiwanese as the low-resource target language.
- **Evaluation Protocol**: We evaluate the model at three stages: before adaptation, after naive or parameter-efficient adaptation, and after RL optimization. This protocol allows us to directly measure emotional forgetting, emotional recovery, and trade-offs with intelligibility and speaker consistency.
- **Standard Evaluation Sets**: For standard synthesis quality and stability testing, we additionally report metrics on benchmark sets such as SeedTTS test-zh, SeedTTS test-en, and LibriSpeech-test-clean when applicable.

4 Methodology

Our methodology follows a two-stage framework built upon a pre-trained expressive IndexTTS2 model. Starting from a model trained on a high-resource language, we first adapt it to the low-resource target language, and then apply reinforcement learning to preserve and recover emotional expressiveness, prosodic control, and content accuracy during transfer.

4.1 Stage 1: Low-Resource Language Adaptation

In the first stage, we adapt the expressive base model to the low-resource target language. The goal of this stage is to acquire target-language pronunciation and linguistic patterns while preserving as much of the pretrained expressive prior as possible. To reduce severe catastrophic forgetting, we adopt a parameter-efficient adaptation strategy, such as lightweight adapter modules or LoRA, so that only a limited subset of model parameters is updated during target-language transfer.

4.2 Stage 2: RL-Based Emotional Realignment

In the second stage, we introduce an RL-based alignment procedure on top of the adapted model. The core idea is to treat the autoregressive generation of semantic speech representations as a sequential decision-making process, while computing task-specific rewards from the decoded speech waveform. This stage explicitly optimizes emotion preservation, emotional intensity, local emphasis, and content intelligibility in the low-resource language. In practice, the RL updates are mainly applied to the autoregressive T2S transformer, and can optionally include the emotion- or style-related conditioning modules when stronger expressive realignment is needed.

4.3 Reinforcement Learning Formulation

The generation of semantic speech tokens is formulated as a Markov Decision Process (MDP), while the reward is computed from the decoded speech waveform:

- **State (S_t):** The hidden states of the IndexTTS2 autoregressive decoder, incorporating the input text, speaker conditions, and previously generated speech tokens up to step t.
- **Action (A_t):** The predicted semantic speech token at step t, sampled from the policy distribution $\pi_\theta(A_t|S_t)$.
- **Reward (R_t):** A composite reward signal provided upon the completion of a sequence or segment. The reward is designed to preserve emotional expressiveness while maintaining intelligibility in the low-resource target language, and comprises four components:
  1. *Emotion Classification Reward:* Evaluated using a pre-trained emotion recognition model (e.g., emotion2vec) to measure whether the generated speech matches the target emotion category.

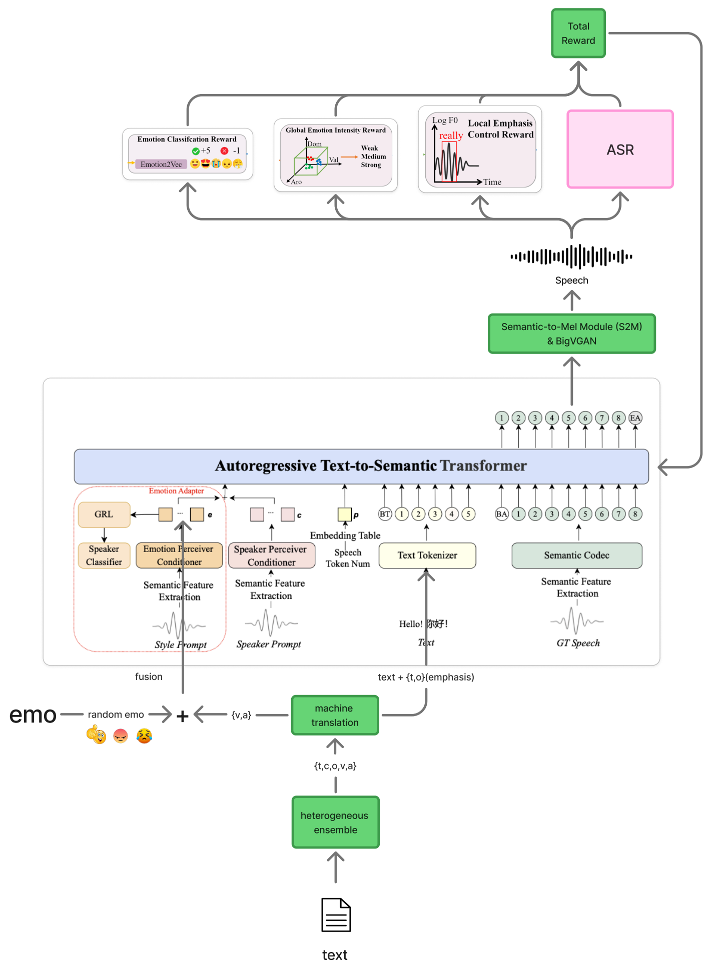

Figure 1: High-level architecture of the proposed RL-enhanced IndexTTS2 framework. After low-resource adaptation, the generator outputs semantic speech tokens, which are decoded into speech waveforms and evaluated by emotion, prosody, and content reward models. These rewards are used to update the policy via Group Relative Policy Optimization (GRPO), mitigating emotional forgetting during language transfer.

2. *Global Emotion Intensity Reward:* Computed from the distance between the generated speech and a neutral centroid in the Valence-Arousal-Dominance (VAD) space, in order to encourage the target global emotional intensity.
3. *Local Emphasis Control Reward:* A reward based on the relative pitch and energy patterns of emphasized words, which encourages correct placement of local stress and expressive prosody.
4. *ASR-Based Content Reward:* Computed using an ASR model on the generated speech to compare the transcription against the input text, encouraging content preservation and intelligibility during RL optimization.

In summary, the sequence-level reward can be written as

$$R = \lambda_1 R_{\text{emo-cls}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}},$$

where the coefficients λ_i control the trade-off between emotional expressiveness, local prosody, and content accuracy.

4.4 Alleviating Catastrophic Forgetting

To address the degradation of emotional capabilities during low-resource fine-tuning, we initialize our policy π_θ with the weights of the target-language adapted IndexTTS2 model rather than training from scratch. We then employ Group Relative Policy Optimization (GRPO) mixed with a Kullback-Leibler (KL) divergence penalty against the adapted reference model. This ensures that the model explores localized emotional and prosodic improvements while preserving the linguistic adaptation and expressive prior inherited from the original high-resource model.

5 Preliminary Results

5.1 Preliminary Baseline Setup

As an initial pilot study, we conducted a subjective evaluation to establish a preliminary baseline for Taiwanese TTS. The evaluation used the Mean Opinion Score (MOS) on a scale from 1 to 5 (where 5 represents the best performance). The evaluation panel consisted of 12 members from the SARC laboratory. The test corpus involved 20 sentences for evaluating speech intelligibility and naturalness, and an additional 10 sentences specifically for evaluating voice similarity. In total, the subjective evaluation yielded 600 rating samples ((20 * 2 + 10) test items $\times$ 12 evaluators).

5.2 Preliminary Baseline Observation

Table 1 shows the preliminary subjective evaluation results of the baseline method, CosyVoice2, on Taiwanese TTS. The model achieves a similarity MOS of 4.16, an intelligibility MOS of 4.41, and a naturalness MOS of 4.37, resulting in an overall mean MOS of 4.31. These results suggest that a strong off-the-shelf baseline can already generate high-quality Taiwanese speech. This observation motivates our project to move beyond basic intelligibility and naturalness, and to focus on whether emotional expressiveness can be preserved and recovered during low-resource adaptation.

Table 1: Preliminary Subjective Evaluation (MOS) on Taiwanese TTS

Model	**Similarity MOS ↑**	**Intelligibility MOS ↑**	**Naturalness MOS ↑**	**Mean ↑**
CosyVoice2	4.16	4.41	4.37	4.31

6 Expected Contributions of Each Team Member

- Hao-Chun Hsieh : 34%
- Pao-Hua Hsu : 33%
- Nien-Tse Wu : 33%

References

Siyi Zhou, Yiquan Zhou, Yi He, Xun Zhou, Jinchao Wang, Wei Deng, and Jingchen Shu. Indextts2: A breakthrough in emotionally expressive and duration-controlled auto-regressive zero-shot text-to-speech. *ArXiv*, abs/2506.21619, 2025. URL `https://api.semanticscholar.org/CorpusID:280010717`.

Ziyang Ma, Zhisheng Zheng, Jiaxin Ye, Jinchao Li, Zhifu Gao, Shiliang Zhang, and Xie Chen. emotion2vec: Self-supervised pre-training for speech emotion representation. *ArXiv*, abs/2312.15185, 2023. URL `https://api.semanticscholar.org/CorpusID:266551115`.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International Conference on Machine Learning*, 2022. URL `https://api.semanticscholar.org/CorpusID:252923993`.

Zhihao Du, Yuxuan Wang, Qian Chen, Xian Shi, Xiang Lv, Tianyu Zhao, Zhifu Gao, Yexin Yang, Changfeng Gao, Hui Wang, Fan Yu, Huadai Liu, Zhengyan Sheng, Yue Gu, Chong Deng, Wen Wang, Shiliang Zhang, Zhijie Yan, and Jing-Ru Zhou. Cosyvoice 2: Scalable streaming speech synthesis with large language models. *ArXiv*, abs/2412.10117, 2024. URL `https://api.semanticscholar.org/CorpusID:274762932`.